INCH-POUND
MIL-DTL-23053/10D
17 June 1996
SUPERSEDING
MIL-I-23053/10C
01 May 1986

DETAIL SPECIFICATION SHEET

INSULATION SLEEVING, ELECTRICAL, HEAT SHRINKABLE, SILICONE RUBBER, FLEXIBLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the sleeving described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-DTL-23053.

REQUIREMENTS:

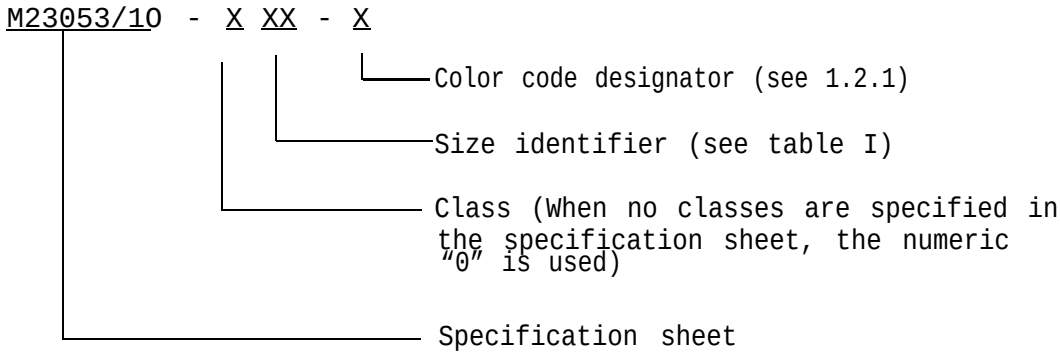
Polymer type: The base elastomer used in formulating this sleeving shall be a silicone.

Continuous operating temperature range: -75°C (-103°F) to +175°C (+347°F).

Color: The sleeving shall be furnished in a black color that conforms to the requirements of Class II of MIL-STD-104.

Longitudinal change: +3, -10 percent

Military part number: The Military part number shall consist of the basic number of this specification sheet and dash numbers shown as follows:



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Example: Black, 2.000 inch (50.80 mm) as supplied ID sleeving shall be identified as M23053/10-012-O.

TABLE I. Construction details, inches (mm). 1/

Military part number 3/	As supplied ID minimum	After unrestricted shrinkage	
		ID maximum	Wall thickness 2/
M23053/10-OO1-0	.125(3.18)	.071(1.80)	.020 ± .006(.508 ± .152)
M23053/10-O02-0	.250(6.35)	.143(3.63)	.035 ± .010(.889 ± .254)
M23053/10-O03-0	.375(9.53)	.214(5.44)	.040 ± .O10(1.O16 ± .254)
M23053/10-O04-0	.500(12.70)	.286(7.27)	.048 ± .015(1.219 ± .381)
M23053/10-O05-0	.625(15.88)	.357(9.07)	.052 ± .015(1.321 ± .381)
M23053/10-O06-0	.750(19.05)	.428(10.87)	.057 ± .015(1.448 ± .381)
M23053/10-O07-0	.875(22.23)	.500(12.70)	.065 ± .015(1.651 ± .381)
M23053/10-O08-0	1.000(25.40)	.570(14.48)	.070 ± .020(1.778 ± .508)
M23053/10-O09-0	1.250(31.75)	.714(18.14)	.087 ± .O2O(2.210 ± .508)
M23053/10-010-O	1.500(38.10)	.857(21.77)	.095 ± .020(2.413 ± .508)
M23053/10-011-O	1.750(44.45)	1.000(25.40)	.107 ± .020(2.718 ± .508)
M23053/10-O12-0	2.000(50.80)	1.140(28.96)	.110 ± .020(2.794 ± .508)

1/ Diameter limits for the object to be enclosed shall be as recommended in technical data.

2/ Wall thickness values are less when shrinkage is restricted.

3/ The color code identified is the standard color.

Unrestricted shrinkage: Test method 4.6.5.2 200° ± 2°C (392° ± 4°F) for 10 minutes.

TABLE II. Physical properties.

Characteristic	Requirement	Test procedure and conditions
As supplied:		
ID, minimum	Table I	4.6.3
Heat shock	No cracks, flowing or dripping	4.6.8 300° ± 4°C (572° ± 7.2°F) No mandrel bend
Tensile stress at 100 percent, psi (MPa), maximum	1,000(6.9)	4.6.12.2 ASTM D412

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TABLE II. Physical properties. - Continued

Characteristic	Requirement	Test procedure and conditions
Cold impact	No cracking	4.6.7.2 -75° ± 2°C (-105° ± 4°F) longitudinal strip.ASTM D746
Dielectric strength, volts/roil (Kv/mm), minimum	200(7.9)	4.6.2 ASTM D2671
Restricted shrinkage	No cracks	4.6.6.1.1 200° ± 2°C (392° ± 4°F)
Voltage withstand	Pass	4.6.6.2
After unrestricted shrinkage:		
ID, maximum	Table I	4.6.3
Wall thickness	Table I	4.6.3
Tensile strength, psi (MPa), minimum	600(4.2)	4.6.13 ASTM D638, 20 inches/minute
Ultimate elongation, percent, minimum	200	4.6.13 ASTM D638, 20 inches/minute
Volume resistivity, Ohm-cm, minimum	1×10^{11}	4.6.2 ASTM D876
Water absorption, Percent, maximum	1.0	4.6.2 ASTM D570, 24 hrs at 23°C
Corrosion	No corrosion or pitting	4.6.10.1 and 4.6.10.2, 175° ± 2°C (347° ± 4°F),16 hours
Heat resistance, properties after:		4.6.9 200° ± 3°C (392° ± 4°F), 168 hours
Tensile strength, psi (MPa), minimum	450 (3.1)	

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TABLE II. Physical properties. - Continued

Characteristic	Requirement	Test procedure and conditions
Ultimate elongation percent, minimum	100	
Fluid resistance, properties after:		4.6.11
Tensile strength, psi (MPa), minimum	300 (2.1)	
Ultimate elongation, percent, minimum	100	
Flammability	Self-extinguishing 1 minute, 25 percent maximum of indicator flag burned	4.6.14 Procedure B ASTM D2671
Fungus resistance	No Growth	4.6.2 ASTM G21

Storage life conditions: Supplier shall certify to storage at 18° to 35°C (65° to 95°F) for one year. Conformance to 3.5. See 3.5.2 for storage life extension.

Intended use: Heat shrinkable, flexible, silicone sleeving is intended for use in high or low temperature applications or in ablative applications.

Custodians:
Army - ER
Navy - AS
Air Force - 85

Review activities:
Army - AR, MI
Navy - EC, SH, MC
Air Force - 99
DLA - GS

Preparing activity:
Navy - AS

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